



# SYNDRONE

GENERATIVE HARMONIC DRONE INSTRUMENT

## Comprehensive User Guide

Version 1.0.5 • Slowfield Audio

*VST3 • Audio Unit • Standalone*

**What Syndrone is** A long-form generative drone synthesizer built around autonomous voices, MIDI interaction, multi-engine synthesis, minute-scale evolution, deep modulation and immersive stereo movement.

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# 1. Getting started

Syndrone is designed to reward slow listening. You can use it as a completely autonomous drone generator, as a MIDI-playable instrument, or as a hybrid in which played notes sit inside an evolving autonomous field.

## 60-second quick start

- Load a factory preset and make sure Perpetual is enabled.
- Choose a Key and Scale, then set Octave Min and Octave Max to define the register.
- Choose a Structure for the synthesis identity and a Character for its overall personality.
- Raise Density for more autonomous voices, then adjust Gravity to decide how strongly the voice field is pulled toward the tonal centre.
- Use Space, Shimmer and Grain Cloud for atmosphere.
- Raise Evolution Amount and choose an Evolution Time of 8 or 16 minutes for long-form movement.
- Use Mass for scale and weight; enable Supermassive when you want the largest possible field.
- Keep Output conservative when experimenting with heavy feedback, saturation or Wild Randomise.

**TIP** A good first move is to select a preset, listen for 30-60 seconds, then change only one major dimension: Structure, Character, Scale or Evolution Shape. Syndrone can change dramatically from a single high-level choice.

## Signal-flow concept

The exact DSP is more interconnected than a traditional left-to-right subtractive synth, but the musical workflow can be thought of as:

| Stage                      | What it does                                                                                       |
|----------------------------|----------------------------------------------------------------------------------------------------|
| Constraints                | Key, scale, octave range and duration define the musical boundaries.                               |
| Voice generation           | Autonomous and MIDI voices create the harmonic source field.                                       |
| Structure + Character      | Choose the synthesis identity and broad spectral personality.                                      |
| Macro shaping              | Density, Gravity, tone-shaping, shimmer, grain, Mass and Age alter the global feel.                |
| Evolution + modulation     | Minute-scale scenes and four deep modulation routes create movement.                               |
| Organism + band processing | Cross-band coupling, filters, sub reinforcement, degradation and polymetric cycles add complexity. |
| Spatial + mixer            | Position and balance Base, Middle, Top and Air before final output.                                |

# 2. How Syndrone thinks

Syndrone is not primarily a note-sequencing synthesizer. It is a field instrument. Rather than programming a phrase and pressing play, you define a tonal world and let multiple systems interact inside it.

## The four time scales

| Time scale | Main systems | Musical effect |
|------------|--------------|----------------|
|------------|--------------|----------------|

|            |                                                      |                                                              |
|------------|------------------------------------------------------|--------------------------------------------------------------|
| Immediate  | Wave, Harmonic, Tone, filters, mixer                 | What the sound is now.                                       |
| Slow       | Drift, Movement, Grain Motion, Spatial Motion        | Continuous animation over seconds.                           |
| Long       | Evolution Engine, harmonic migration, scene morphing | Structural development over minutes.                         |
| Polymetric | Band cycles and 5-stage lane                         | Repeating changes that do not need to align with each other. |

**HOT TIP** Avoid turning every movement system up at once. A strong drone often uses one dominant long-form motion source and two or three subtler supporting motions.

### 3. Interface overview

The fixed 1520 × 1216 interface is divided into ten functional areas. The Roman-numeral section markers are a useful way to think about programming order.

| Section                              | Purpose                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------|
| I - Constraints                      | Key, scale, octave range, Character and Structure.                                  |
| II - Duration                        | Fixed bar lengths or perpetual operation.                                           |
| III - MIDI / Performance             | XY field, MIDI blend, Mutation, Mass and related performance macros.                |
| IV - Spatial Engine                  | Stereo width, depth, height, orbit, distance and motion.                            |
| V - Evolution Engine                 | Minute-scale movement, scenes and harmonic migration.                               |
| VI - Macro / Atmosphere              | Density, Gravity, Space, Shimmer, Grain and FX Freeze.                              |
| VII - Synthesis Engine               | Wave, Harmonic, Movement, Spread, Evolve, Voices, Flavor, Drift, Coupling and Tone. |
| VIII - Deep Modulation / Band Timbre | Envelope, four modulation routes, dual band filters and colour.                     |
| IX - Organism / Degradation          | Saturation, bit/rate reduction, morph voices, topologies and polymetric lane.       |
| X - Mixer                            | Base, Middle, Top and Air level balance.                                            |

### 4. Pitch, scale and duration

#### Key and Scale

Key sets the tonal centre. Scale constrains both autonomous generation and MIDI quantisation. Syndrone includes conventional modes and more unusual pitch collections.

| Scale family       | Available scales                                                    |
|--------------------|---------------------------------------------------------------------|
| Common tonal       | Major, Natural Minor, Dorian, Phrygian, Lydian, Mixolydian, Locrian |
| Minor variants     | Harmonic Minor, Melodic Minor, Neapolitan Minor                     |
| Pentatonic / blues | Major Pentatonic, Minor Pentatonic, Blues, Major                    |

|                       |                                                                     |
|-----------------------|---------------------------------------------------------------------|
|                       | Blues                                                               |
| Symmetric / synthetic | Whole Tone, Diminished, Suspended, Prometheus, Enigmatic            |
| World / colour        | Japanese In, Hirajoshi, Phrygian Dominant, Hungarian Minor, Persian |

Octave Min and Octave Max define the autonomous register. Narrow ranges produce more coherent fields; wide ranges create orchestral separation and stronger contrast between Base, Middle and Top.

**TIP** For dark drones, try an octave span of 1-4. For luminous or glassy drones, try 3-7. A wide 1-7 range can be spectacular, but it is more sensitive to Density and Mass.

## Duration

Fixed Length offers 16, 32 or 64 bars. Perpetual ignores the fixed endpoint and allows continuous evolution. Fixed bar calculation follows host tempo and is designed around conventional 4/4 timing.

## 5. Synthesis Engine

| Control  | Function                                                                 | Practical use                                                                                |
|----------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Wave     | Moves through 96 algorithmic spectral variants across multiple families. | Use small moves for colour; large moves can change the identity of a patch.                  |
| Harmonic | Changes harmonic emphasis.                                               | Low settings tend to be simpler; higher settings can produce richer or more complex spectra. |
| Movement | Rate/intensity of spectral animation.                                    | Keep low for static monuments; raise for living textures.                                    |
| Spread   | Detune and stereo dispersion.                                            | Adds width and ensemble character.                                                           |
| Evolve   | Longer-term structural movement inside the synthesis voice.              | Useful even with Evolution Amount at zero.                                                   |
| Voices   | Autonomous voice count.                                                  | More voices increase density and CPU use.                                                    |
| Flavor   | Nonlinear colour inside Structures.                                      | One of the fastest ways to make the same Structure feel different.                           |
| Drift    | Slow oscillator instability.                                             | Excellent for tape-like, organic and aged motion.                                            |
| Coupling | Cross-influence between Base, Middle and Top.                            | Makes the bands behave more like one organism.                                               |
| Tone     | Global spectral tilt.                                                    | Use as the primary bright/dark control before detailed filters.                              |

**HOT TIP** Coupling is more than “more movement”. At moderate settings it can make separate bands feel causally connected. At high settings it can create unstable, almost feedback-like organic behaviour.

## 6. Character and Structure

### 16 Structure modes

| Structure          | Character                                                                 |
|--------------------|---------------------------------------------------------------------------|
| Dual Drift         | Two drifting spectral tendencies; balanced and versatile.                 |
| Opposed Harmonics  | Contrasting harmonic directions for tension and movement.                 |
| Triple Bloom       | Layered harmonic blooming; good for expansive ambient beds.               |
| Ring Atmos         | Ring-like spectral interaction and metallic atmosphere.                   |
| FM Atmos           | Frequency-modulation-inspired colour for glassy or complex drones.        |
| Warm Blend         | Rounded, blended spectra for gentle beds.                                 |
| Fold               | More nonlinear harmonic folding and edge.                                 |
| Phase Cloud        | Diffused phase relationships and hazy movement.                           |
| Subharmonic        | Undertone and low-frequency reinforcement.                                |
| Organismic         | Cross-coupled, unstable and living behaviour.                             |
| Additive Resonator | Harmonic/inharmonic partial banks; bells, resonances and metallic bodies. |
| Formant Choir      | Vowel-like weighting for vocal and choral material.                       |
| Phase Glass        | Bright phase-related spectra and crystalline surfaces.                    |
| Cluster Unison     | Dense micro-detuned clusters for huge stereo fields.                      |
| Bowed Resonance    | Sustained body/resonance character suggestive of bowed material.          |
| Spectral Organ     | Structured harmonic stacks for organ-like and ceremonial drones.          |

**TIP** When auditioning Structures, temporarily reduce Shimmer and Grain Cloud. That exposes the actual oscillator identity so you can hear what the Structure is contributing.

### 16 Character types

| Character  | Tendency                                             |
|------------|------------------------------------------------------|
| Ancient    | Dark, weathered and weighty.                         |
| Cosmic     | Wide, airy and spatial.                              |
| Frozen     | Cold, restrained and crystalline.                    |
| Organic    | Drifting and naturally unstable.                     |
| Sacred     | Balanced, resonant and ceremonial.                   |
| Dust       | Softened, aged and textured.                         |
| Glass      | Clear, bright and resonant.                          |
| Submerged  | Darkened and distant.                                |
| Warm       | Rounded upper spectrum and relaxed motion.           |
| Mechanical | More regular, precise and edged movement.            |
| Hollow     | Reduced central body with resonant/formant emphasis. |
| Dense      | Greater spectral occupancy and weight.               |
| Fragile    | Thin, airy and delicate.                             |

|            |                                                    |
|------------|----------------------------------------------------|
| Monolithic | Fundamental-heavy, slow and massive.               |
| Radiant    | Bright, open and shimmering.                       |
| Alien      | More asymmetric, inharmonic and unusual behaviour. |

**HOT TIP** Structure answers “what kind of synthesis is this?” Character answers “what personality does that synthesis have?” Treat them as two independent axes. The most interesting patches often come from non-obvious combinations.

## 7. Macro / Atmosphere

| Control          | Purpose                                                                     |
|------------------|-----------------------------------------------------------------------------|
| Density          | Population of the autonomous voice field.                                   |
| Gravity          | Pull toward the tonal centre; higher settings produce stronger pitch focus. |
| Space            | Main diffuse reverb size/wetness.                                           |
| Shimmer          | Octave-shifted atmospheric return.                                          |
| Shimmer Feedback | Feedback around the shimmer path.                                           |
| Grain Cloud      | Amount of granular diffusion.                                               |
| Grain Size       | Granular delay-window length.                                               |
| Grain Motion     | Irregular movement in the grain field.                                      |
| FX Freeze        | Holds the atmospheric effect state for sustained texture.                   |
| Age / Patina     | Macro that darkens, destabilises and gently degrades the sound.             |
| Mass             | Adds scale, detuned body and reinforcement with gain compensation.          |
| Supermassive     | Pushes the scale/body concept substantially further.                        |

**HOT TIP** FX Freeze is especially effective after a dramatic harmonic moment. Freeze the atmosphere, then change the underlying synthesis so the frozen tail and new source coexist.

**CAUTION** Supermassive, high Density, heavy Shimmer feedback and high Coupling can create a very energetic patch. Syndrone includes protection and gain compensation, but monitor your output level when combining extremes.

## 8. Evolution Engine and Scenes

The Evolution Engine is designed for changes measured in minutes, not conventional LFO speed. It can reshape harmony, timbre, effects and space while Scenes provide four captured musical states.

| Control          | What it does                                           |
|------------------|--------------------------------------------------------|
| Evolution Amount | Overall depth of minute-scale development.             |
| Evolution Time   | 1, 2, 4, 8, 16 or 32 minutes per cycle.                |
| Evolution Shape  | Rise/Fall, Breathing, Wander, Tide, Bloom, Collapse or |

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
|                    | Generative.                                                           |
| Timbre             | Depth of long-form wave, tone and density development.                |
| Harmony            | Long-form harmonic/scene emphasis.                                    |
| Space              | Amount of evolving stereo/spatial movement.                           |
| FX                 | Independent long-cycle Shimmer/Grain development.                     |
| Harmonic Migration | Changes favoured scale degrees while staying inside the chosen scale. |
| Rare Events        | Smooth occasional blooms, air expansions and harmonic events.         |
| Scene Path         | Order used to travel through A-D.                                     |
| Scene Morph        | Enables continuous travel between captured states.                    |
| Generate Scenes    | Creates a set of contrasting scene states.                            |
| Capture A-D        | Stores the current musical state into each scene.                     |

**TIP** For convincing long-form evolution, start with Amount around 20-35% and an 8-16 minute cycle. If you can obviously hear “the modulation”, the amount may be too high for a background drone.

**TIP** Harmonic Migration is designed to avoid obvious portamento smears: voices fade, retune to a scale degree, then return. Use it for harmonic motion without abandoning the tonal centre.

## 9. Spatial Engine

| Control        | Purpose                                                                    |
|----------------|----------------------------------------------------------------------------|
| Immersive      | Master amount for the psychoacoustic stereo field.                         |
| Width          | Upper-band stereo width while protecting low-frequency mono compatibility. |
| Depth          | Front/back impression using decorrelation and short delays.                |
| Height         | Subtle upper-spectrum elevation cue.                                       |
| Orbit          | Left/right orbital movement.                                               |
| Distance       | Damping and narrowing that moves the sound perceptually away.              |
| Motion         | Speed/complexity of spatial movement.                                      |
| Space Size     | Expands the spatial reverb field.                                          |
| Auto Orbit     | Automatically animates orbital movement.                                   |
| Evolution Link | Lets the Evolution Engine drive spatial change.                            |

**HOT TIP** Width and Orbit are not interchangeable. Width makes the field broader; Orbit makes it move. For a stable cinematic bed, use high Width with low Orbit. For a living headphone texture, do the opposite.

## 10. Deep Modulation

Syndrone provides four modulation slots. Each slot has a Source, Destination and bipolar Amount. Negative amounts invert the source. The routing is smoothed to reduce clicks from stepped/random sources.

### Modulation sources

- Slow LFO A-F
- Hyper LFO
- Envelope
- Random Walk
- Sample & Hold
- Sample & Glide
- Chaos
- Bipolar Random
- Smooth Random
- Brownian Walk
- Evolution
- Scene Morph
- Step Lane
- Host Tempo Phase
- Euclidean Pulse
- Probability Pulse

### Modulation destinations

- Harmonic
- Movement
- Spread
- Evolve
- Tone
- Shimmer
- Grain Cloud
- Coupling
- Tension
- Grain Size
- Grain Motion
- Shimmer Feedback
- Width
- Depth
- Height
- Orbit
- Distance
- Mass
- Age / Patina
- Base Level
- Middle Level
- Top Level
- Air Level

- Base Filter
- Middle Filter
- Top Filter
- Morph A
- Morph B
- Saturation
- Rate Reduce

**HOT TIP** Brownian Walk -> Age / Patina at a small amount can make a clean drone slowly “weather” and recover without sounding periodically modulated.

**HOT TIP** Probability Pulse -> Air Level or Shimmer gives occasional flashes rather than continuous motion. This is useful when you want events, not wobble.

**HOT TIP** Host Tempo Phase -> Grain Motion or Orbit creates movement that relates to the DAW tempo without turning Syndrone into a conventional synced arpeggiator.

## 11. Organism, degradation and polymetric evolution

### Organism / degradation

| Control                 | Function                                                   |
|-------------------------|------------------------------------------------------------|
| Base / Middle / Top Sub | Independent one-octave-down reinforcement per band.        |
| Saturation              | Nonlinear drive before the organismic coupling stage.      |
| Bit Depth               | Moves from clean resolution toward coarse digital texture. |
| Rate Reduce             | Sample-hold-style rate reduction.                          |
| Tension / Fracture      | Adds nonlinear fracture and stronger cross-band tension.   |
| Stereo Asymmetry        | Offsets left/right saturation for animated imbalance.      |
| Bank Spread             | Detune/phase dispersion inside each multi-voice bank.      |
| Morph A / B             | Levels of two additional morphing spectral voices.         |
| A Wave / B Wave         | Spectral position for the two morph voices.                |
| Filter 2                | Second cutoff in each dual-filter band path.               |
| Topology                | Series, Parallel or Split Stereo filter routing.           |
| Bank Voices             | 1, 2 or 3 internal voices per band.                        |

**TIP** Use degradation selectively. A small Rate Reduce with full Bit Depth can add motion without making the patch overtly lo-fi. Age / Patina can then supply the broader aged character.

## Independent band cycles

Base, Middle, Top and Air can each run an independent evolution cycle: Off, 5, 7, 9, 11 or 13 bars. Because these lengths differ, the overall combination may take a long time to repeat.

**HOT TIP** Try Base = 7 bars, Middle = 11 bars, Top = 13 bars, Air = 9 bars. The layers constantly realign, producing complex evolution without randomising the underlying harmony.

## Five-stage evolution lane

The host-synchronised five-stage lane contains Step Cycle, Destination, Amount and stages S1-S5. It can target the same broader destination set used by the expanded modulation system.

**TIP** Use a small negative Step Amount on Tone or Filter to create slow darkening pulses. A negative amount is often more musical than simply reversing stage values.

## 12. Four-layer mixer

Syndrone separates the final tonal field into Base, Middle, Top and Air. These are not simply four oscillators: they represent register and atmospheric components at the end of the synthesis architecture.

| Layer  | Role                                                       |
|--------|------------------------------------------------------------|
| Base   | Foundation, subharmonic weight and low-register body.      |
| Middle | Main harmonic body and recognisable tonal content.         |
| Top    | Upper harmonic detail and brightness.                      |
| Air    | Shimmer/grain atmospheric return and high-field diffusion. |

**TIP** If a patch feels muddy, reduce Middle before cutting Base. Many drones need a strong low foundation but less mid-band occupancy.

## 13. MIDI and performance

MIDI Blend controls the balance of played MIDI voices and autonomous generation. At one extreme, Syndrone behaves more like a self-running instrument; at the other, played notes become a stronger part of the field.

The XY performance pad maps four perceptual directions: Air, Dark, Bright and Fracture. It is intended for broad expressive movement rather than precision parameter editing.

**HOT TIP** Record slow XY gestures into your DAW automation while Evolution continues underneath. This gives you a human performance layer on top of the autonomous system.

## 14. Presets, favourites and Randomise

The factory bank contains 52 presets, including showcase patches and simpler starting points. Presets are tagged by categories such as Ambient, Dark, Choir, Massive, Evolving, Experimental and Starter.

Starter presets include Init Clean Drone, Warm Bed, Glass Air and Deep Monolith. Use these when you want to programme from a clearer baseline.

The star control marks favourite factory presets. User presets are stored as .syndronepreset files and can be saved/loaded from the Slowfield Audio/Syndrone Presets folder in your Documents directory.

### Randomise

| Mode     | Behaviour                                                      |
|----------|----------------------------------------------------------------|
| Gentle   | Preserves more of the current patch and makes smaller changes. |
| Medium   | Broad exploration while remaining reasonably controlled.       |
| Wild     | Radical parameter movement intended to find unexpected states. |
| Mutation | Controls the depth of the selected Randomise mode.             |

**TIP** Use Gentle at low Mutation as a “variation” button. Use Wild only after saving a sound you want to keep.

## 15. Drone recipes

The recipes below are starting points. Values are approximate percentages; the purpose is to illustrate relationships between controls rather than prescribe exact numbers.

### Beginner recipe 1 - Warm cinematic bed

| Area                  | Setting                                |
|-----------------------|----------------------------------------|
| Key / Scale           | D / Dorian                             |
| Octaves               | 2-5                                    |
| Structure / Character | Warm Blend / Warm                      |
| Density / Gravity     | 45% / 70%                              |
| Movement / Drift      | 25% / 30%                              |
| Space / Shimmer       | 55% / 12%                              |
| Evolution             | Amount 20%, 8 min, Breathing           |
| Spatial               | Width 65%, Depth 45%, Orbit 10%        |
| Mixer                 | Base 55%, Middle 65%, Top 35%, Air 25% |

Result: stable, musical and forgiving. Increase Tone for more clarity or Mass for more cinematic weight.

### Beginner recipe 2 - Frozen glass halo

| Area                  | Setting              |
|-----------------------|----------------------|
| Key / Scale           | A / Major Pentatonic |
| Octaves               | 3-7                  |
| Structure / Character | Phase Glass / Frozen |
| Density / Gravity     | 30% / 55%            |

|                       |                                     |
|-----------------------|-------------------------------------|
| Wave / Harmonic       | 65% / 70%                           |
| Spread / Tone         | 70% / 78%                           |
| Shimmer / Grain Cloud | 38% / 18%                           |
| Evolution             | Amount 15%, 16 min, Tide            |
| Spatial               | Width 80%, Height 55%, Distance 35% |

**TIP** If Phase Glass becomes brittle, lower Tone before reducing Shimmer. That preserves the spatial halo while softening the source.

### Beginner recipe 3 - Deep monolith

| Area                  | Setting                                |
|-----------------------|----------------------------------------|
| Key / Scale           | C / Natural Minor                      |
| Octaves               | 1-4                                    |
| Structure / Character | Subharmonic / Monolithic               |
| Density / Gravity     | 35% / 85%                              |
| Base Sub / Mid Sub    | 60% / 25%                              |
| Mass / Supermassive   | 70% / Off initially                    |
| Movement / Drift      | 12% / 18%                              |
| Evolution             | Amount 10%, 16 min, Rise/Fall          |
| Mixer                 | Base 80%, Middle 45%, Top 15%, Air 10% |

### Intermediate recipe 1 - Living organism

| Area                  | Setting                               |
|-----------------------|---------------------------------------|
| Key / Scale           | E / Phrygian                          |
| Structure / Character | Organismic / Organic                  |
| Coupling / Tension    | 65% / 35%                             |
| Bank Spread           | 45%                                   |
| Band cycles           | Base 7, Middle 11, Top 13, Air 9 bars |
| Evolution             | Amount 35%, 8 min, Generative         |
| Harmonic Migration    | 25%                                   |
| Rare Events           | 18%                                   |
| Mod 1                 | Brownian Walk -> Coupling, +20%       |
| Mod 2                 | Smooth Random -> Tone, -12%           |

### Intermediate recipe 2 - Choir in a ruined hall

| Area                  | Setting                                         |
|-----------------------|-------------------------------------------------|
| Key / Scale           | F# / Harmonic Minor                             |
| Structure / Character | Formant Choir / Ancient                         |
| Octaves               | 2-6                                             |
| Shimmer / Space       | 22% / 70%                                       |
| Age / Patina          | 42%                                             |
| Saturation            | 18%                                             |
| Evolution             | Amount 30%, 16 min, Bloom                       |
| Scene Morph           | On; capture A dark, B bright, C sparse, D dense |
| Mod 1                 | Evolution -> Shimmer, +18%                      |
| Mod 2                 | Scene Morph -> Middle Level, +15%               |

### Intermediate recipe 3 - Metallic orbital field

| Area                  | Setting                                 |
|-----------------------|-----------------------------------------|
| Key / Scale           | B / Whole Tone                          |
| Structure / Character | Additive Resonator / Mechanical         |
| Spread / Flavor       | 55% / 70%                               |
| Tension               | 30%                                     |
| Spatial               | Immersive 75%, Orbit 60%, Auto Orbit On |
| Evolution             | Amount 22%, 8 min, Wander               |
| Mod 1                 | Slow LFO E -> Harmonic, +20%            |
| Mod 2                 | Host Tempo Phase -> Orbit, +22%         |
| Step lane             | 8 bars -> Grain Motion, 25%             |

### Intermediate recipe 4 - Dusty memory

| Area                    | Setting                             |
|-------------------------|-------------------------------------|
| Key / Scale             | G / Minor Pentatonic                |
| Structure / Character   | Dual Drift / Dust                   |
| Age / Patina            | 55%                                 |
| Drift                   | 48%                                 |
| Bit Depth / Rate Reduce | 85% / 10%                           |
| Space / Grain           | 52% / 28%                           |
| Evolution               | Amount 18%, 32 min, Breathing       |
| Mod 1                   | Brownian Walk -> Age / Patina, +15% |
| Mod 2                   | Sample & Glide -> Top Filter, -10%  |

### Advanced recipe 1 - Polymetric super-field

| Area                  | Setting                              |
|-----------------------|--------------------------------------|
| Key / Scale           | C# / Enigmatic                       |
| Structure / Character | Cluster Unison / Cosmic              |
| Density / Voices      | High / high but CPU-aware            |
| Mass / Supermassive   | 78% / On                             |
| Band cycles           | 5 / 7 / 11 / 13 bars                 |
| Evolution             | Amount 38%, 16 min, Generative       |
| Scene Path            | A-C-B-D                              |
| Mod 1                 | Brownian Walk -> Width, +18%         |
| Mod 2                 | Probability Pulse -> Air Level, +30% |
| Mod 3                 | Slow LFO F -> Grain Size, -20%       |
| Mod 4                 | Evolution -> Mass, +12%              |

**CAUTION** Cluster Unison + Supermassive + high voice count + heavy shimmer/grain is one of the most CPU-intensive combinations. Reduce Voices or Density first if the DAW crackles.

### Advanced recipe 2 - Alien harmonic weather

| Area        | Setting        |
|-------------|----------------|
| Key / Scale | E / Prometheus |

|                       |                                  |
|-----------------------|----------------------------------|
| Structure / Character | Phase Glass / Alien              |
| Harmonic / Flavor     | 82% / 70%                        |
| Coupling / Tension    | 50% / 48%                        |
| Rare Events           | 32%                              |
| Evolution             | Amount 45%, 8 min, Collapse      |
| Mod 1                 | Chaos -> Harmonic, +22%          |
| Mod 2                 | Euclidean Pulse -> Morph A, +30% |
| Mod 3                 | Bipolar Random -> Distance, +16% |
| Mod 4                 | Scene Morph -> Tone, -20%        |

### Advanced recipe 3 - Bowed ritual machine

| Area                  | Setting                                  |
|-----------------------|------------------------------------------|
| Key / Scale           | D / Hungarian Minor                      |
| Structure / Character | Bowed Resonance / Sacred                 |
| Movement / Drift      | 38% / 30%                                |
| Coupling / Saturation | 45% / 22%                                |
| Base / Mid Sub        | 35% / 18%                                |
| Evolution             | Amount 30%, 16 min, Tide                 |
| Step lane             | 4 bars -> Tension, stages 20/65/35/80/25 |
| Mod 1                 | Envelope -> Coupling, +12%               |
| Mod 2                 | Random Walk -> Top Filter, +18%          |

### Advanced recipe 4 - Spectral cathedral

| Area                  | Setting                               |
|-----------------------|---------------------------------------|
| Key / Scale           | A / Lydian                            |
| Structure / Character | Spectral Organ / Radiant              |
| Octaves               | 2-7                                   |
| Harmonic / Tone       | 68% / 74%                             |
| Space / Shimmer       | 72% / 38%                             |
| Evolution             | Amount 28%, 32 min, Bloom             |
| Scenes                | A sparse; B full; C dark; D radiant   |
| Mod 1                 | Scene Morph -> Shimmer Feedback, +15% |
| Mod 2                 | Slow LFO A -> Height, +20%            |
| Mod 3                 | Brownian Walk -> Middle Filter, +12%  |

### Advanced recipe 5 - Submerged transmission

| Area                  | Setting                              |
|-----------------------|--------------------------------------|
| Key / Scale           | Bb / Japanese In                     |
| Structure / Character | Ring Atmos / Submerged               |
| Octaves               | 1-5                                  |
| Tone / Top Filter     | 35% / 28%                            |
| Age / Patina          | 38%                                  |
| Distance / Depth      | 70% / 75%                            |
| Grain Motion          | 45%                                  |
| Evolution             | Amount 25%, 16 min, Wander           |
| Mod 1                 | Sample & Hold -> Grain Motion, +18%  |
| Mod 2                 | Probability Pulse -> Air Level, +20% |
| Mod 3                 | Evolution -> Distance, +12%          |

## 16. Hot tips

### Sound-design shortcuts

- Change Structure before reaching for EQ-like controls. It produces a more fundamental transformation.
- Use Character to establish the emotional identity, then Tone and band filters for refinement.
- For drones that feel alive rather than busy, keep one modulation route obvious and the others subtle.
- Use Gravity as a coherence control: high Gravity makes the tonal centre obvious; lower Gravity lets the field wander.
- Use Morph A/B as additional spectral voices, not just “more level”. Different A/B Wave positions can create a second harmonic identity inside the patch.
- Use Split Stereo topology on Top while keeping Base more centred for width without weakening the foundation.
- Age / Patina and Dust Character are complementary but not identical. Use both lightly rather than maxing one.
- If a patch sounds washed out, reduce Air Mix before reducing Space. You may want the room but not as much atmospheric return.
- Use Rare Events sparingly. Their strength comes from being rare.
- For endless installations or sleep/meditation beds, use Perpetual + 16/32-minute Evolution + modest Rare Events.

### Performance shortcuts

- Save before Wild Randomise.
- Capture Scene A before performing major changes so you always have a stable anchor.
- Use XY gestures for macro expression and the modulation matrix for repeatable internal movement.
- Automate MIDI Blend to move between self-playing and performer-led passages.
- If you want a dramatic transition without a hard preset change, morph between Scenes rather than loading another preset.

## 17. Troubleshooting and CPU management

| Symptom                | First things to try                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------|
| Clicks/crackles        | Reduce Voices or Density; disable Supermassive; lower Grain/Shimmer; increase DAW buffer size.         |
| Patch too loud         | Lower Output; reduce Mass/Supermassive; reduce Saturation/Tension; compare with a simpler preset.      |
| Patch too muddy        | Reduce Middle Mix; reduce Coupling; brighten band filters; reduce Space/Air Mix.                       |
| Patch too bright/harsh | Lower Tone; reduce Top Mix; choose a warmer Character; reduce Shimmer Feedback.                        |
| Too much movement      | Lower Evolution Amount, Movement, Spatial Motion and modulation amounts before changing the Structure. |
| Too static             | Raise Evolve or Evolution Amount; add one slow modulation route; enable Evolution Link or Auto Orbit.  |

|                       |                                                                              |
|-----------------------|------------------------------------------------------------------------------|
| MIDI feels buried     | Raise MIDI Blend and reduce autonomous Density.                              |
| DAW cannot see plugin | Rescan plugins; verify VST3/AU installation path; restart the DAW if needed. |

**NOTE** Syndrone includes smoothing, gain ramps, DC blocking, feedback limits, output protection and CPU-aware voice limits. Very heavy combinations can still exceed a system or DAW buffer budget, which sounds like crackling even when the synthesis itself is stable.

## 18. Installation notes and release checklist

### macOS

The release package is intended to install VST3, Audio Unit and Standalone formats. The current beta packaging is not code-signed/notarised, so macOS Gatekeeper may require an explicit Open action. Final commercial distribution should use Developer ID signing and notarisation.

### Windows

The Windows installer is intended to install the VST3 and Standalone application. Unsigned beta installers may trigger Microsoft Defender SmartScreen. Final commercial distribution should use an Authenticode code-signing certificate.

### Recommended pre-release validation

- Build and install on a clean macOS machine or test account.
- Validate VST3 and AU discovery in Ableton Live and Logic Pro.
- Build Windows x64 through GitHub Actions and test the installer on a clean Windows system.
- Save and reopen DAW projects with multiple Syndrone instances to verify state recall.
- Run long 30-60 minute Perpetual sessions to check stability and memory behaviour.
- Audition all 52 factory presets for level consistency and obvious duplication.
- Stress-test Cluster Unison + Supermassive + high Density + heavy Grain/Shimmer.
- Confirm user preset save/load and favourite recall.

**Release philosophy** At this point the most valuable work is testing, installer reliability and preset polish rather than adding new features. Freeze the feature set once v1.0.5 passes build and DAW validation.